

Unit 3 : Session 12 : SLO 2 — Tutorial: IEEE 754 Double Precision Encoding

[illegible]

- Roles: Decimal Converter, Normalizer, Exponent Encoder (bias 1023, 11-bit), Sign Bit Decider, Mantissa Extractor (52-bit fraction), Encoder Assembler (final 64-bit word).
- Execution: each student completes their stage and passes the result on; the final student reads out the full 64-bit binary.
- Discussion: if the number is too large for the exponent range it overflows to infinity; if too small it underflows toward denormalized numbers or zero — precision is limited by the 52-bit mantissa.